HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented automated test scripts using **Python**, improving regression test coverage by **40%** for critical manufacturing systems.
- Collaborated with cross-functional teams, including Quality and Manufacturing Engineering, to analyze device failures and identify root causes for **implantable electronic medical devices**.
- Performed device failure analysis on **electronic medical devices**, meticulously documenting findings and contributing to component-level root cause identification.
- Developed and executed validation activities for test equipment and manufacturing processes, ensuring compliance with **medical device standards** and internal procedures.
- Supported software developers in defining and implementing system enhancements that improved manufacturing and **test yields** for electronic components.
- Prepared clear and accurate **technical documentation** and supported regulatory submissions through established change notice processes.
- Analyzed manufacturing data using **SQL** and internal tracking systems to identify trends, troubleshoot issues, and support continuous improvement initiatives.

SKILLS

Programming Languages – Python, C, C++, SQL, Java, R

Tools & Technologies – LabVIEW, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace, Docker, Linux, AWS

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn, R/RStudio

Domain Knowledge – Medical Device Manufacturing, Electronic Manufacturing Systems, Test Fixtures, Product Quality, Lean Manufacturing, Regulatory Compliance, GDPR

Methodologies – Agile, Scrum, Change Management, Technical Documentation, Data Analysis, Root Cause Analysis, Validation

PROJECTS

Automated Test Fixture Validation Suite

- Developed a **Python**-based framework to automate the validation of custom test fixtures used in medical device manufacturing, reducing manual validation time by **60%**.
- Integrated test fixture performance data into a centralized dashboard using **Power BI** for real-time monitoring and trend analysis by the engineering team.
- Documented the test suite architecture, usage guidelines, and validation procedures, supporting knowledge transfer and adoption across the test engineering team.

Manufacturing Process Data Analysis and Reporting

- Engineered **Python** scripts to extract, transform, and load manufacturing process data from various sources, including sensor logs and internal databases.
- Applied statistical methods and **data analysis** techniques to identify key performance indicators and anomalies in production lines, contributing to a **15%** reduction in process fallout.
- Built interactive dashboards in **Power BI** to visualize process performance, enabling manufacturing engineers to make data-informed decisions for quality improvements.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)